Local & Global Extrema: Which Test, When?

Name _____

START: find the critical numbers
Solve $f'(x) = 0$ and find where $f'(x)$ does not exist. Keep only x-values in the domain of f.

What is the question asking for?

LOCAL MAX / MIN

Is $f'(c) = 0$ and $f''(c)$ easy to find?

YES → 2nd Derivative Test

- $f''(c) > 0$ → local **MIN** (concave up)
- $f''(c) < 0$ → local **MAX** (concave down)
- $f''(c) = 0$ → **inconclusive**, so use the 1st test

▼ if $f''(c) = 0$

NO (f' DNE or f'' messy) → 1st Derivative Test

Sign chart of f' on each side of c:

- **+ to −** → local **MAX**
- **− to +** → local **MIN**
- **same sign** → neither

GLOBAL (ABSOLUTE) MAX / MIN

What is the domain?

CLOSED $[a, b]$ → Closed Interval Method

- Evaluate f at every critical number **inside $[a, b]$ and** at both endpoints (f continuous). An extremum can sit at an endpoint.
- Largest value → global **MAX**
- Smallest value → global **MIN**

OPEN or UNBOUNDED ((a, b) or $\mathbb{R}$) → Critical Numbers + End Behavior

1. Find $f(c)$ at each critical number in the domain (and at an endpoint only if it is **included**).
2. Check each **open or infinite end**: the limit of f, or the sign chart of f'.
3. Global **MAX** = the biggest $f(c)$ that no end beats ($f \to \infty$, or a larger limit never reached). Global **MIN**: same idea, lowest.

Key details

CRITICAL NUMBERS

c is critical if it is in the domain of f and $f'(c) = 0$ **or** $f'(c)$ does not exist (a corner, cusp, or vertical tangent). They are only **candidates**.

TRAP: CRITICAL ≠ EXTREMUM

$f(x) = x^3$ has $f'(0) = 0$ but no max or min. And $f''(c) = 0$ does not mean "no extremum": x^4 has a min at 0, x^3 has none. Go to the sign chart.

SAY IT IN WORDS (1ST DERIVATIVE TEST)

"f' changes from + to − at $x = c$, so f has a local max at $x = c$." Test one x in each interval of the number line; it works at every critical number, even where f' DNE.

OPEN OR UNBOUNDED SHORTCUT

Only **one** critical number c on the whole domain? f' goes **− to +** → global **min** at c; **+ to −** → global **max** at c. The other extreme usually does not exist. *e.g.* $x^2 - 4x$ on $(-\infty, \infty)$: global min $f(2) = -4$, and $f \to \infty$ both ways, so no global max.

WORKED EXAMPLE $f(x) = x^3 - 12x$

1. $f'(x) = 3x^2 - 12 = 3(x - 2)(x + 2)$, defined everywhere, so critical numbers are $x = -2$ and $x = 2$.
2. **Local:** $f''(x) = 6x$. $f''(-2) = -12 < 0$ → local **max**, value $f(-2) = 16$. $f''(2) = 12 > 0$ → local **min**, value $f(2) = -16$.
3. **Global on $[-3, 5]$:** compare the critical numbers **and** the endpoints (table at right).
4. **Global on $(-\infty, \infty)$:** none. $f \to \infty$ as $x \to \infty$ and $f \to -\infty$ as $x \to -\infty$.



sign of f' (1st derivative test agrees: max at −2, min at 2)

x	−3 *(end)*	−2	2	5 *(end)*
$f(x)$	9	16	**−16**	**65**

Global max **65** at $x = 5$ (an endpoint, not the local max).
Global min **−16** at $x = 2$.